What is new inside SafeQuant:

The original version of SafeQuant can be found at https://github.com/eahrne/SafeQuant.

Below you will find all the modifications to the newly updated SQ script, which can be found at https://github.com/georgiaAngelidou/SafeQuant.v2.4.

New Experimental Design:

1. You can set the experimental design by using a .txt file instead of specifying it on the command line. This allows you to define the labels for each condition more conveniently.

Changes to the default settings:

1. For MaxQuant files, if the user does not include **–SR**, raw intensities will be used. If **–SR** is included, the LFQ values will be used.
2. By default, the program calculates the mean value for each biological condition rather than the median.
3. The default imputation method has been changed to nDist (imputation based on the normal distribution of the data).
4. By default all the exported areas are transformed to log2.
5. The columns **FTestPValue**, **FTestQValue**, and those starting with f**racNAFeatures** have been removed from the default output table. These columns can be included in the output only by using the **–extraOut** flag**.**

New feature:

1. The **nDist** method is now the default imputation method and is automatically applied unless a different method is specified using the **–SM** flag.
2. **–Median:** When this flag is used, the program will use median values for further statistical calculations instead of the default setting, which is the mean.
3. –**noPDF:** This flag can be used to generate only the output table and exclude the PDF report.
4. **–log2to**l: By default, all values are transformed to log2. To use raw or untransformed values, include this flag in your command.
5. **–extraOut:** This flag will include output columns that were present in the original version of SQ but are hidden in the current version.
6. –**s SelectedProteinsList.csv:** If you have specific proteins you'd like to check separately, use this flag. You'll need to provide a .csv file containing the gene names of the proteins of interest, and trend plots will be generated for those targets.
7. A PCA plot has been included to the PDF report.
8. A violin plot comparing normalized and non-normalized data has been added to the PDF report.
9. Three new bar plots have been added to the PDF report, displaying the following information:
    a. Total number of precursors measured per file
    b. Total number of proteins measured per file
    c. Total number of imputed values per file

10. The script supports DIANN files after the DIANN-output is converted via the script "DIANN-SafeQuant-report_convert_protein_20230807.R"